

House Price Prediction Using Machine Learning

Project Summary

This project focused on predicting **house prices** using machine learning techniques based on various property characteristics such as **area, number of bedrooms, bathrooms, stories, parking facilities, air conditioning, and furnishing status**. The dataset was obtained from Kaggle and was first explored, cleaned, and preprocessed before building predictive models. Categorical features were converted into numerical form using **one-hot encoding**, and the dataset was divided into **80% training data** and **20% testing data** for model evaluation.

Two regression models were developed: **Linear Regression** and **Random Forest Regressor**. The models were evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. These metrics helped measure the accuracy of the predictions and compare the performance of the two models. The **Random Forest Regressor** achieved better results than Linear Regression, indicating that it was more effective at capturing the complex relationships between house features and their prices.

The analysis revealed that **area** was the most influential factor affecting house prices. Other important features included **bathrooms, parking availability, air conditioning, and number of stories**. Houses with larger areas and better amenities generally had significantly higher prices. The correlation heatmap further confirmed that these features had a strong positive relationship with house prices.

One interesting finding was that some features, such as **guest rooms** and **hot water heating**, had a smaller impact on house prices than expected. This suggests that buyers place greater importance on the overall size of the property and essential facilities rather than certain additional amenities.

In simple terms, the machine learning model was able to predict house prices with **good accuracy** and relatively low prediction error. The results demonstrate that machine learning can be a valuable tool for estimating property values and supporting data-driven decision-making in the real estate sector.

Recommendation

Based on the findings, real estate businesses should focus on **property area, parking facilities, bathroom count**, and other key amenities when pricing properties. Using machine learning-based price estimation systems can improve pricing accuracy, reduce reliance on guesswork, and help both buyers and sellers make more informed decisions.